

RESEARCH-BASED VIRTUAL PRACTICAL: DNA EXTRACTION FROM STRAWBERRY



Student:

Hamid Nawaz

Internship:

Biotechnology Internship

Organization:

Code Alpha

Task 3:

Laboratory/Practical Task

Practical Type:

Research-Based / Virtual

Date:

September 2026

Table of Contents

1. AIM	v
2. OBJECTIVES.....	v
3. INTRODUCTION.....	v
4. PRINCIPLE	vi
5. MATERIALS REQUIRED	vi
6. PROCEDURE.....	viii
7. EXPERIMENTAL WORKFLOW	xi
8. OBSERVATIONS.....	xi
9. RESULTS	xii
10. ROLE OF EACH COMPONENT	xiii
11. DISCUSSION.....	xiv
12. APPLICATIONS OF DNA EXTRACTION	xv
13. LIMITATIONS	xvi
14. SAFETY PRECAUTIONS	xvi
15. CONCLUSION	xvii
17. REFERENCES	xviii

List of Table

Table 1 Materials Requires for practical	vii
Table 2 Observation	xi
Table 3 Data Show Results of Exraction.....	xiii

List of Figures

Figure 1 Strawberry + cells + DNA ka conceptual visual.....	vi
Figure 2 Materials Required	vii
Figure 3 Visual representation of DNA Extraction from Strawberry.....	x
Figure 4 Observation.....	xii
Figure 5 Role of Components in Strawberry	xiv
Figure 6 Principles of DNA Extraction.....	xv
Figure 7 Application of DNA Extraction	xvi

1. AIM

To demonstrate the extraction and visualization of DNA from strawberry tissue using a simple research-based DNA extraction procedure involving mechanical disruption, detergent-mediated cell lysis, filtration, and cold-alcohol precipitation.

2. OBJECTIVES

After completing this virtual/research-based practical, the learner should be able to:

- i. Understand the basic principle of DNA extraction.
- ii. Explain why fruit tissue is mechanically disrupted.
- iii. Understand the role of detergent and salt in DNA extraction.
- iv. Explain how alcohol precipitates DNA.
- v. Recognize the expected visual appearance of extracted DNA.
- vi. Relate simple DNA extraction to professional molecular-biology techniques.

Fruit DNA extraction activities from NHGRI and South Dakota State University use essentially these principles and materials.

3. INTRODUCTION

Deoxyribonucleic acid, commonly referred to by its abbreviation DNA, constitutes the fundamental molecule that serves the essential function of storing and preserving the genetic information upon which all living organisms depend for their existence and biological continuity. In the case of plant cells, it is within the nucleus that the predominant portion of the cell's DNA is located. Prior to the application of any laboratory technique—whether it be the polymerase chain reaction (PCR), DNA sequencing, gel electrophoresis, or genetic analysis—for the purpose of studying DNA, it is first necessary to release the DNA from within the cells and to isolate it from the remaining cellular constituents. Consequently, the process of DNA extraction constitutes one of the most essential and foundational procedures within the disciplines of molecular biology and biotechnology. A simple and accessible experiment that employs fruit as its starting material in order to extract DNA is capable of demonstrating the very same fundamental principles that serve as the basis for the considerably more sophisticated DNA-isolation procedures performed in professional research laboratories.

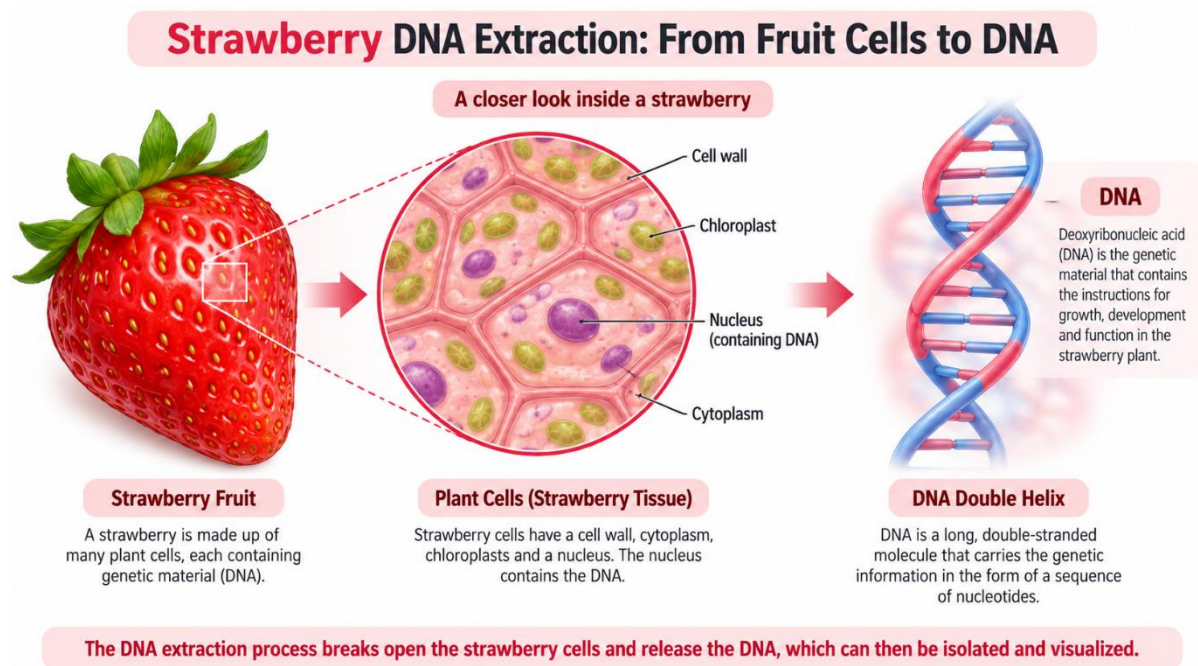


Figure 1 Strawberry + cells + DNA ka conceptual visual.

4. PRINCIPLE

The extraction is based on four major processes:

i. Mechanical disruption

The strawberry is crushed to break down the plant tissue and expose its cells.

ii. Cell lysis

Dishwashing detergent disrupts the lipid membranes of the cells and nuclei, releasing DNA into the solution.

iii. Salt-assisted precipitation

Salt helps neutralize the negative charge associated with DNA and promotes aggregation of DNA molecules.

iv. Alcohol precipitation

DNA has poor solubility in cold alcohol. When chilled ethanol or isopropanol is added, DNA precipitates and becomes visible as a whitish/stringy material.

5. MATERIALS REQUIRED

For the practical demonstration:

No.	Material	Purpose
1	2 strawberries	DNA source
2	Resealable plastic bag	Crushing sample
3	Water	Extraction medium
4	Dishwashing detergent	Cell lysis
5	Table salt	Helps DNA aggregation
6	Coffee filter/sieve	Removes debris
7	Plastic cups/beaker	Mixing and collection
8	Chilled rubbing alcohol/ethanol/isopropanol	DNA precipitation
9	Spoon/stirrer	Mixing
10	Toothpick/glass rod	Collecting DNA

Table 1 Materials Requires for practical

NHGRI provides a very similar strawberry protocol using two strawberries, detergent, salt, water, a coffee filter and cold rubbing alcohol.



Figure 2 Materials Required

6. PROCEDURE

Step 1 Prepare the strawberry

Remove the green leaves from approximately two strawberries.

Place the strawberries into a resealable plastic bag.

Step 2 Crush the sample

Seal the bag and gently crush the strawberries for approximately **2 minutes** until a relatively uniform pulp is produced.

Step 3 Prepare extraction solution

In a separate container, prepare:

- ½ cup water
- 2 teaspoons dishwashing detergent
- 1 teaspoon salt

Mix gently until the salt dissolves. Avoid excessive foam.

This composition is based on the NHGRI educational protocol.

Step 4 Lyse the cells

Add the extraction solution to the crushed strawberries.

Seal the bag and gently massage the mixture for approximately **1 minute**.

The detergent disrupts cellular and nuclear membranes and releases DNA into the liquid.

Step 5 Filter

Place a coffee filter over a clean cup.

Pour the strawberry mixture through the filter.

The filter retains the larger pieces of fruit while allowing the DNA-containing liquid to pass through.

Step 6 Add cold alcohol

Transfer some of the filtered liquid into a clean transparent container.

Carefully add approximately an equal volume of **ice-cold alcohol** without vigorously mixing the two layers.

Step 7 Observe

Allow the mixture to stand.

A **whitish, cloudy or string-like material** should appear around the boundary between the strawberry extract and alcohol layer.

This represents precipitated DNA-containing material.

Step 8 Collect the DNA

A clean toothpick or glass rod can be used to gently spool/collect the visible material.



How to extract DNA from a strawberry

Cells are the basic unit of life and make up all plants, animals and bacteria. Deoxyribonucleic acid, or DNA, is the molecule that controls everything that happens in the cell. This activity will demonstrate how DNA can be isolated from a strawberry using common household materials.



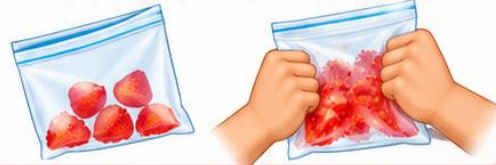
What you will need:



1 Pull off any green leaves on the strawberry.



2 Put the strawberries into the plastic bag, seal it and gently smash the strawberries for about two minutes. This starts to break open the cells and release the DNA.



3 Mix together 2 teaspoons of detergent, 1 teaspoon of salt and 1/2 cup of water in a plastic cup. This will be your DNA extraction liquid.



4 Pour the DNA extraction liquid into the bag with the strawberries (make sure the bag is big enough; quart-sized works well!). This will further break open the cells.



5 Reseal the bag and gently smash for another minute. Avoid making too many soap bubbles.



6 Place the coffee filter inside the other plastic cup. Open the bag and pour the strawberry liquid into the filter. Twist the filter just above the liquid and gently squeeze the remaining liquid into the cup.



7 Pour an equal amount of rubbing alcohol as there is strawberry liquid down the side of the cup. Do not mix or stir.



8 Within a few seconds, watch for a white cloudy substance (DNA) to develop in the top layer. You have just isolated the DNA from the rest of the material contained in the cells of the strawberry.



9 Tilt the cup and pick up the DNA using a coffee stirrer.



Figure 3 Visual representation of DNA Extraction from Strawberry

7. EXPERIMENTAL WORKFLOW

Strawberry
↓
Mechanical Crushing
↓
Detergent+Salt+Water
↓
Cell Lysis
↓
Filtration
↓
Cold Alcohol Addition
↓
DNA Precipitation
↓
Visible White/Stringy DNA

8. OBSERVATIONS

Important: Since we are documenting this as a **research-based/virtual practical**, these are **expected observations**, not fake personal measurements.

Stage	Expected Observation	Interpretation
Crushing	Strawberry becomes soft pulp	Tissue is mechanically disrupted
Extraction solution added	Mixture becomes liquid/cloudy	Cellular contents are released
Detergent treatment	Foaming/cloudiness may occur	Membrane lipids are disrupted
Filtration	Liquid becomes relatively clearer	Large fruit debris is removed
Cold alcohol added	Separate liquid layer forms	Alcohol is less dense and poorly mixes
Interface	White/cloudy/stringy material appears	DNA precipitates
Collection	String-like material may stick to rod	Aggregated DNA becomes visible

Table 2 Observation

NHGRI and UQ both describe formation of visible whitish/string-like material after alcohol treatment.

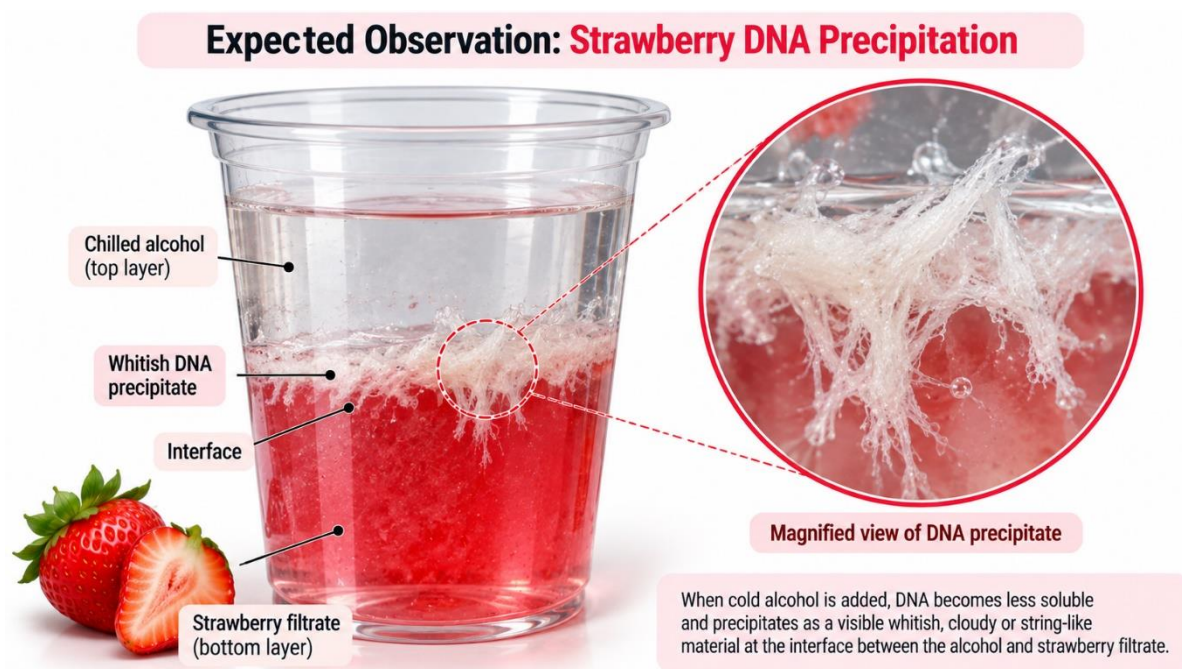


Figure 4 Observation

9. RESULTS

Research-Based Result

The research-based virtual analysis indicates that successful fruit DNA extraction should result in the formation of a **visible whitish or string-like precipitate at the interface between the aqueous strawberry extract and cold alcohol.**

This visual material represents precipitated DNA together with potentially other cellular material.

Because this is a qualitative educational extraction, **DNA concentration and purity cannot be determined from visual appearance alone.**

For professional laboratory analysis, DNA concentration and quality would require analytical methods such as spectrophotometry or gel electrophoresis.

Result Statement

The expected outcome of the research-based practical was the formation of visible whitish, thread-like DNA precipitate following treatment of filtered strawberry extract with chilled alcohol.

Stage	Expected Observation	Interpretation
Crushing	Strawberry becomes pulp	Tissue disruption
Filtration	Relatively clearer liquid	Debris removal
Alcohol addition	Two layers form	Phase separation
Interface	White/cloudy material	DNA precipitation
Collection	String-like material	Aggregated DNA

Table 3 Data Show Results of Extraction

10. ROLE OF EACH COMPONENT

Strawberry

Provides plant cells containing genomic DNA.

Detergent

Breaks down lipid membranes surrounding cells and nuclei.

Salt

Helps reduce electrostatic repulsion between DNA molecules and promotes aggregation.

Water

Provides the aqueous environment for cellular components.

Cold alcohol

Reduces DNA solubility and causes DNA to precipitate.

These roles are supported by institutional educational protocols.

Role of Components in Strawberry DNA Extraction

Each component plays an important role in releasing, separating and visualizing DNA.

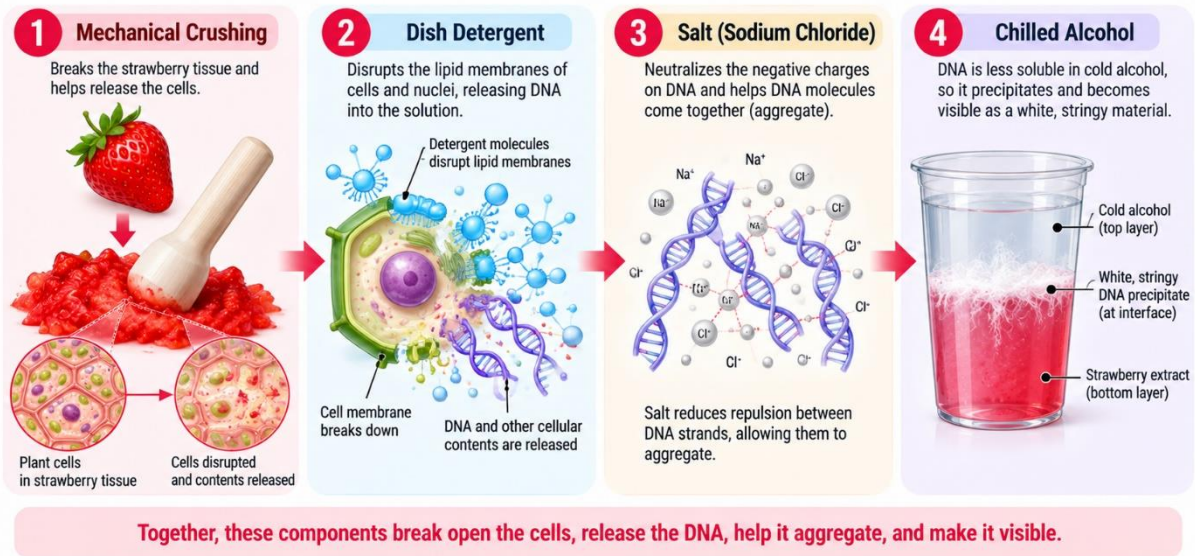


Figure 5 Role of Components in Strawberry

11. DISCUSSION

This practical exercise serves to illustrate the essential and foundational stages involved in the process of DNA extraction, accomplished through the use of readily available and simple materials. The initial phase of this procedure consists of mechanical disruption, a step in which the fruit tissue is broken apart into progressively smaller fragments, thereby facilitating the exposure of the individual cells contained within it. Subsequently, the detergent acts upon the lipid membranes that surround both the cells and their nuclei, disrupting these structures and thereby causing the DNA to be released into the extraction solution. The addition of salt serves to promote the aggregation of DNA molecules, whereas the process of filtration is employed to eliminate the larger fragments of cellular debris from the mixture. In the final stage, the application of cold alcohol reduces the solubility of the DNA to a considerable degree, causing it to precipitate out of the aqueous solution. The DNA that becomes visible as a result of this procedure serves to demonstrate a significant principle of molecular biology: namely, that while individual DNA molecules are not ordinarily observable to the naked eye, substantial quantities of DNA are capable of aggregating together to form a mass that can indeed be seen. This straightforward method employing fruit is intended primarily for educational purposes. It does not yield DNA of the degree of purity that would be anticipated from a professional extraction performed within the field of molecular biology, given that proteins, RNA, pigments, and various other cellular components are liable to remain within the resulting preparation.

Principle of DNA Extraction from Strawberry Tissue

From cells to visible DNA: a simple but powerful process

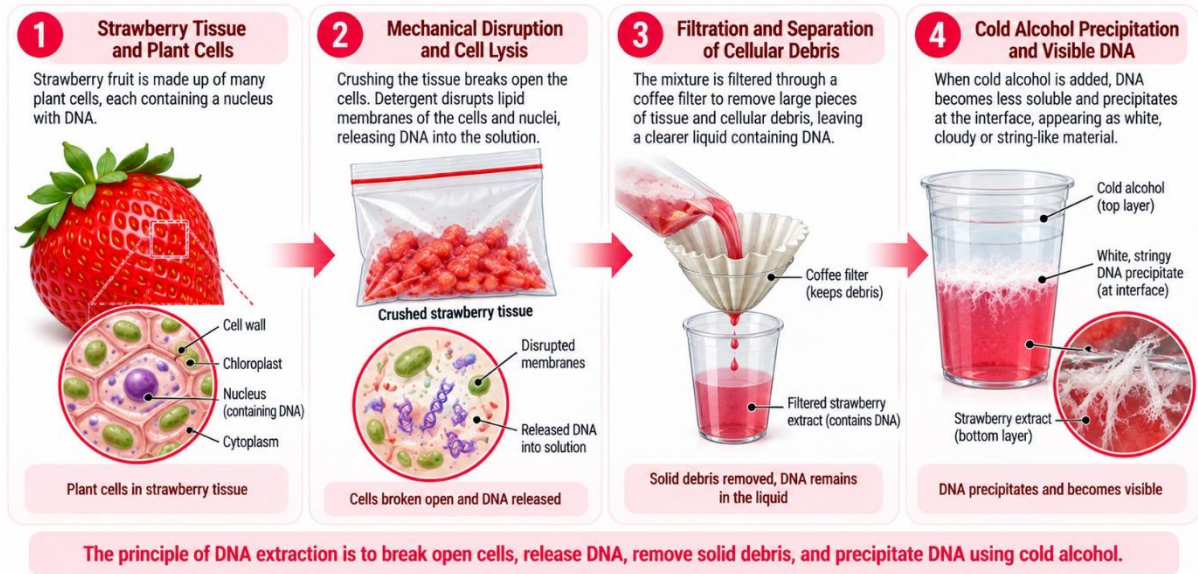


Figure 6 Principles of DNA Extraction

12. APPLICATIONS OF DNA EXTRACTION

DNA extraction is an important first step in many biotechnology applications, including:

- PCR
- DNA sequencing
- Gel electrophoresis
- Genetic testing
- Forensic analysis
- Genetic engineering
- Molecular diagnostics
- Genomic research

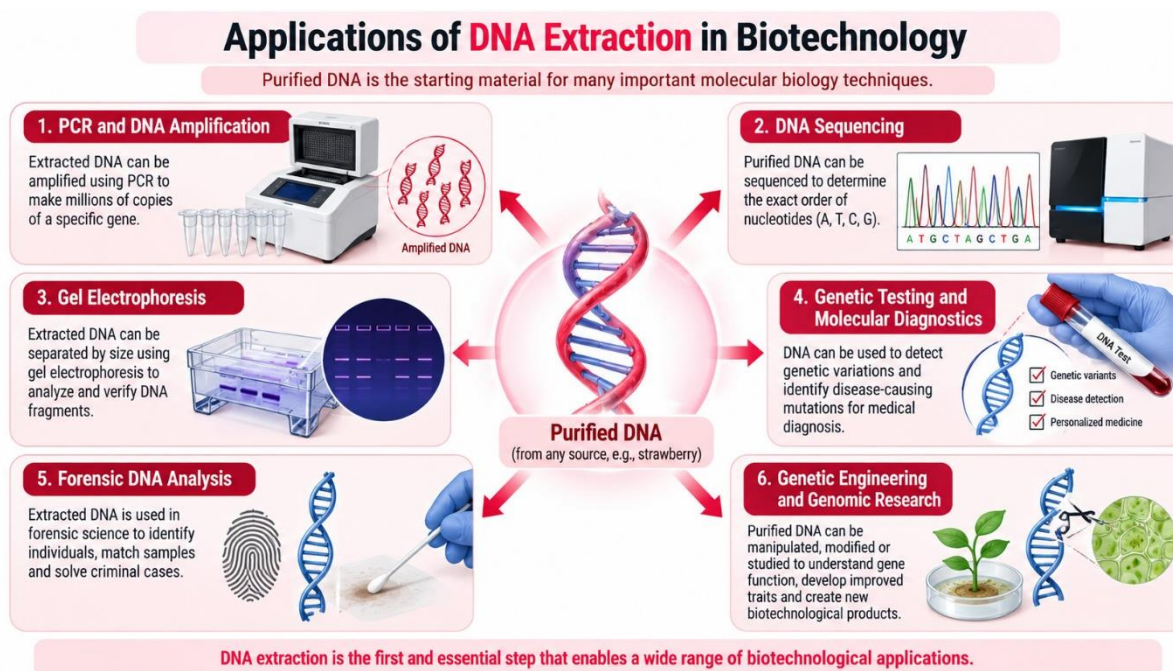


Figure 7 Application of DNA Extraction

13. LIMITATIONS

1. This is a qualitative rather than quantitative experiment.
2. Visible precipitate does not confirm DNA purity.
3. DNA concentration cannot be calculated from visual observation.
4. Other cellular molecules may be present with the DNA.
5. No spectrophotometer or gel electrophoresis was used.
6. The practical is documented as a **research-based/virtual activity rather than a personally performed laboratory experiment.**

14. SAFETY PRECAUTIONS

- Wear gloves when appropriate.
- Avoid contact of detergent or alcohol with eyes.
- Do not consume any materials used in the experiment.
- Handle alcohol away from flames and ignition sources.
- Use clean equipment.
- Keep the workspace clean.
- Dispose of biological/food waste appropriately.

15. CONCLUSION

The virtual practical exercise, which was grounded in prior research, served to illustrate the core principles that underlie the process of extracting DNA from the tissue of strawberries. The breakdown of the fruit tissue is accomplished through mechanical disruption; the cellular and nuclear membranes are then compromised by the action of a detergent; the aggregation of DNA is promoted by the presence of salt; and, finally, the DNA is precipitated by cold alcohol, which causes it to become visible to the observer. This activity offers a straightforward demonstration of a technique that is fundamental to the field of molecular biology, and it further serves to illustrate the manner in which the basic principles of DNA isolation are connected to more advanced applications in biotechnology, including the polymerase chain reaction (PCR), DNA sequencing, genetic engineering, and molecular diagnostics. Taken as a whole, the practical exercise effectively demonstrates the theoretical process by which DNA is isolated and subsequently visualized, while simultaneously underscoring the important point that professional-level DNA analysis necessitates the use of additional purification procedures as well as rigorous quality-control techniques.

17. REFERENCES

National Human Genome Research Institute. (2023). *Strawberry DNA extraction*. National Institutes of Health.

Pourier, A., Pomarleau, W., Nelson, D., Cutler, K., & Nepal, M. P. (2022). *Teaching DNA extractions from fruits (strawberry, banana, kiwi, and watermelon)*. iLEARN Teaching Resources, 4(1), 61–68. South Dakota State University.

YourGenome. (2024). *Extracting DNA from fruit: Activity guide*. Wellcome Genome Campus.

University of Queensland. (n.d.). *Strawberry DNA extraction activity*. Institute for Molecular Bioscience.

University of Minnesota. (n.d.). *Strawberry DNA extraction*. Science for All Outreach Program.